

Get In Touch with Your Seat: Accessible Seat Localization for Blind and Low Vision Travelers in Autonomous Shuttles

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Abstract

A persistent and vexing challenge for blind and low vision (BLV) travelers involves finding assigned and/or available seating in different transportation modalities. This is particularly problematic as autonomous vehicle (AV) rideshares and shuttles become more prevalent given the absence of assistance from human operators. While audio-based smartphone systems are the current standard for BLV interaction, emerging evidence suggests that multisensory systems, combining haptics and spatialized audio, may improve accessibility across the trip. We present a user study with 8 BLV participants comparing 1) audio-only descriptions and 2) multisensory information using vibro-audio maps and environmental audio cues to assist in seat finding. Results suggest higher accuracy, faster completion times, and stronger user preference for the multisensory condition. Participants also emphasized the importance of well-timed, contextually relevant information and greater control over information delivery. These findings highlight the promise of multisensory systems to improve independent travel for BLV users in future AVs.

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CCS Concepts

• **Human-centered computing** → Accessibility; Accessibility technologies; • **Hardware** → Communication hardware, interfaces and storage; Tactile and hand-based interfaces; Haptic devices.

Keywords

Blind and Low Vision Users, Automated Vehicles, Accessibility

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1 Introduction

Shared mobility in autonomous vehicles (AVs) has the potential to provide safe and affordable travel for millions of blind and low vision (BLV) users who currently depend on human drivers. Unlocking this potential necessitates providing accessible information across every stage of the AV trip [6, 11]. While researchers have explored new smartphone-based accessibility tools for finding and navigating to AVs [11], as well as using dedicated haptic devices to assist BLV users exiting AVs [18], there is a dearth of available research exploring the equally important entering and seat localization portion of the trip. Perhaps most relevant to the goals of this paper, recent work by Fink et al. sought to identify a consistent framework for providing audio-based natural language (NL)



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descriptions across each stage of the trip with AVs [9]. While the authors identified a ‘description logic’ for the entering and seat localization phase, they did not experimentally test the identified logic to measure its usability or impact on user performance. Moreover, the authors called for future work to compare their audio-only approach to multisensory interfaces given recent results emphasizing the utility of multisensory information in AVs for BLV users [13, 17, 18]. Given the potential advantages of multisensory access and the need for more user testing of multisensory transportation-related interfaces, the present work seeks to evaluate the NL description logic identified by Fink et al. [9] in a comprehensive user study with and without multisensory cues using two conditions: information with NL descriptions only (the traditional unimodal approach) and information with additional multisensory cues, as described and justified in the following.

AV accessibility projects have leveraged myriad techniques including haptic access [7, 17, 18], augmented visual information [11], spatialized audio [5, 10], and the latest class of multisensory maps that combine vibration and audio, known as vibro-audio maps (VAMs)[20] to enhance situation awareness and improve task completion during AV trips [13]. In this work, we add to this small but growing body of literature by applying our prototype solution that combines VAMs (haptics, audio, and enhanced visual cues) with spatialized audio cues in the environment. In a user study with ($N = 8$) BLV users, we compare our combined multisensory approach with the traditional audio-only technique, focused specifically on supporting users to safely enter and find a target seat in a multipassenger AV simulator. Results suggest that the multisensory condition is preferred by participants and may improve performance, particularly during the entry phase of the trip. Key contributions include a first-of-its-kind prototype VAM applied to a seat localization task, results challenging the assumption that audio-only information is preferred by BLV travelers, as well as important insight on nonvisual mapping and multisensory user interface design in AVs.

2 Related Work

2.1 Blind and Low Vision Challenges Across the Trip with AVs

Although the majority of accessibility focused BLV projects have primarily concerned the in-vehicle stage of AV travel [2, 3, 7, 10, 12, 16, 17], several recent projects have also explored solutions to challenges for other segments of the trip. For instance, researchers have identified navigating to and identifying the correct AV as a significant challenge for BLV travelers and developed a computer-vision based smartphone app that both identified obstacles on the way to the vehicle and the correct door handle [11]. Finding a preferred seat can also be difficult for BLV people [1, 14] and will be particularly relevant in the entry phase of the trip given shared mobility concepts for AVs that include multi-passenger shuttles (e.g., at airports). Finally, exiting AVs safely and orienting to the destination has been a challenge identified in prior work that developed a novel haptic device to disambiguate the arrival environment [18]. Through participatory design methods, researchers have also identified information important for each of these trip segments and proposed a progression and ordering of this information for

future AV deployments, which they dub a ‘description logic’ [9]. While commendable, the prior work derived results from interviews and surveys with participants but did not experimentally test the logic through formal behavioral studies (as we seek to do here). Moreover, the project relied exclusively on audio-based descriptions instead of leveraging the body of work suggesting the advantages of multisensory VAMs, as reviewed in the following.

2.2 Vibro-audio Maps for Multisensory AV Access

A promising approach for achieving multisensory interaction for BLV users in the transportation context is through VAMs. Leveraging design guidelines for smartphone-based implementation [20], VAMs use the phone’s native vibration motor and audio output capabilities to render oriented haptic lines with audio information that together have been shown to aid in cognitive map development [19]. For the tasks studied here (i.e., safely entering a vehicle and navigating to a seat), we posit that VAMs are uniquely positioned to develop the requisite cognitive maps to complete these tasks on a form factor (i.e., smartphones) known to be desirable and nearly ubiquitous among BLV users [12]. Indeed, recent work has shown that VAMs have improved spatial confidence and reasoning for BLV users in tracking and navigating to rideshared vehicles over unimodal, audio-only, interfaces [13]. Given these promising results, we combine VAMs with spatialized audio cues in the environment, as emerging results suggest that spatial audio and haptics are the ideal multisensory combination for BLV travelers when using AVs [5].

3 Methods

The study leveraged a description logic informed by previous work [9] to test two conditions, NL only and a multisensory condition. The study followed a within-subjects methodology, counterbalanced to avoid learning effects. 8 self-identified legally blind participants, aged 23-54 ($M = 37.63$, $SD = 11.98$), were recruited for the study across a range of visual status and mobility aid usage (Appendix 1). The research was approved by a university IRB and participants were compensated with \$50 cash and a meal.

3.1 Apparatus and Procedure

Participants were individually tasked with entering the cabin of an autonomous vehicle simulator (Figure 1 left) and locating their assigned seat. The AV simulator was elevated approximately one-foot above ground level and accessed via a two-step platform with a handrail for assistance. The interior consisted of 6 seats arranged in a semicircle opening towards the door, representing “campfire seating” common to current shuttle vehicles.

The study employed a Wizard-of-Oz approach where one iPhone, controlled by the research team, delivered seat information to the participant’s iPhone. This seat information was paired with additional information BLV individuals have previously identified as important to include in entry description logics (e.g., safety information in the vehicle, climate control location, and traffic information) [8, 9]. In both the multisensory and NL conditions, audio descriptions were presented before and after entering the simulator based on the ordering suggested in [9].



Figure 1: (Left) The AV simulator used for seat localization and (Right) the vibro-audio map used in the study

Table 1: Vibration paramaters and audio cues used in seat-localization VAMs

Audio Cue	Pattern	Intensity	Duration	Interval
“3 o’clock bench”	Constant	50%	1s	1Hz (continuous vibration)
“12 o’clock bench”	Constant	50%	1s	1Hz (continuous vibration)
“9 o’clock bench”	Constant	50%	1s	1Hz (continuous vibration)
“Your seat”	Pulse	100%	.05s / .2s	4Hz
“Entrance”	Pulse	100%	.15s	~7Hz

The NL condition gave audio-only spatial descriptions to orient and instruct participants and was designed to resemble traditional audio-only interfaces that many BLV participants are familiar with (e.g., VoiceOver, TalkBack, JAWS, etc.), while also including spatial language like “at 11 o’clock” or “to the left,” commonly used in orientation and mobility training [10]. Audio descriptions before entering the simulator gave information such as: “You will be sitting on the bench to your right at 3 o’clock in the left-hand seat because the other seat has been flagged for cleaning. The vehicle’s destination is Aroma Joe’s. Please be mindful of the step inside as you enter the vehicle and locate your seat.” The seat location was changed (left vs. right) between conditions.

The multisensory condition leveraged a prototype VAM (Figure 1 right) developed using guidelines from prior work [19, 20], the same NL descriptions, as well as sonification via a chime that could be elicited by the user from the target seat to convey relevant spatial and semantic information. The VAM used the iPhone’s onboard vibration motor and unique vibration patterns for the entrance, seating layout, and target seat (Table 1).

A short training session with the VAM was conducted to ensure participants had an appropriate baseline understanding of how to use the display. During training, participants used the VAM to find the entrance point, trace along the vibrating lines which represented the seat benches, and locate the point indicating the seat. Following training, participants completed a set of criterion questions and all participants passed the criterion tests. In both conditions, participants were instructed to focus on the information delivered, as they would be asked questions and give feedback about the quality, amount, and timing. A clock was started to measure task completion time after participants heard the audio and

experienced the VAM. Participants could request that information be repeated within each condition and the number of repeated cues was recorded by the research team.

3.2 Post-Test Questions

Following each condition, structured audio-recorded interviews were conducted with participants using an 8-question post-condition questionnaire to gather feedback on clarity, helpfulness, and timing of the information (Appendix 2). These feedback questions were first rated on 5-point Likert scales to evaluate their agreement (1, Strongly Disagree to 5, Strongly Agree), and then followed by similar, open-ended questions to further gather participants’ opinions. Participants were also asked ten questions using the System Usability Scale (SUS) to quantify their opinions on the information used in both the NL and multisensory conditions.

3.3 Analysis

Statistical analyses were conducted using R and JASP with a significance level of $\alpha = .05$. Participant completion times were analyzed using a paired sample t-test and the Likert scale ratings were performed with a Wilcoxon signed-rank test as a non-parametric alternative. SUS scores for both conditions were compared using a paired sample t-test to assess significant differences in perceived usability. While these statistical tests were employed, we recognize their fragility given the low sample size. Additionally, the average SUS scores for each condition were compared against industry-standard SUS percentiles created by Lewis and Sauro [15] to contextualize their usability relative to a broader benchmark. Lastly,

Table 2: Summary results comparing Accuracy, Time, and SUS Scores for the multisensory and natural language conditions

Accuracy (N, %)			Time (s)				SUS Scores			
<i>NL</i>	<i>Multisensory</i>	<i>p</i>	<i>NL</i>	<i>Multisensory</i>	<i>p</i>	<i>d</i>	<i>NL</i>	<i>Multisensory</i>	<i>p</i>	<i>d</i>
4/8 (50%)	8/8 (100%)	NA	52.38	18.13	.04*	.89	82.19 (A)	87.5 (A+)	.19	.51

inferential statistics were not employed for participant task accuracy due to the study's small sample size and discreteness of the accuracy measurement.

4 Results

In the multisensory condition, all 8 participants (100%) identified the correct seat, whereas only 4 out of 8 (50%) identified the correct seat in the NL condition. However, the 4 participants who incorrectly identified the seat still sat on the correct side of the AV simulator. Participants took almost 3 times longer, on average, to complete the NL condition (52.38 seconds, $SD = 37.07s$), compared to the multisensory condition (18.13 seconds, $SD = 13.73s$). A paired sample t-test indicated a statistically significant difference in task completion time ($t(7) = 2.53$, $p = .04$, $d = .89$). This was likely due to 4 participants requesting audio instructions be repeated within the NL condition, whereas only 2 participants requested additional repeats within the multisensory condition, including an additional 3 participants repeating the seat chime cue. 7 out of 8 participants (87.5%) preferred the multisensory condition over the NL condition, however, opinions diverged on specific features. For instance, while the option to trigger a seat-location sound was generally appreciated, some participants thought it was unnecessary for confirmation.

As part of the 8-question post-condition questionnaire, when participants were asked if the information presented was helpful, the Likert median score for the NL condition was 4.5 ($IQR = 1.0$), while the multisensory condition was 5.0 ($IQR = 1.0$) [$W = 4.0$, $p = .77$, $r = .19$]. When asked if the information was presented at appropriate times, the Likert median score for the NL condition was 4.5 ($IQR = 1.0$), while for the multisensory condition was 4.0 ($IQR = 1.0$) [$W = 0$, $p = .35$, $r = .47$]. The Likert median score for the appropriate amount of information for the NL condition was 4.0 ($IQR = 0.5$), while for the multisensory condition was 4.5 ($IQR = 1.0$) [$W = 6.0$, $p = .15$, $r = .57$]. The average SUS score for the NL condition was 82.19 ($SD = 14.60$), while the average SUS score for the multisensory condition was 87.50 ($SD = 9.64$), with this difference not being reliably different $t(7) = 1.45$, $p = .19$, $d = .51$. According to the Lewis and Sauro benchmarking [15], the NL condition was graded "A", while the multisensory condition was graded "A+". Table 2 summarizes these results.

In long answer responses, participant feedback on features they liked centered around the clarity and confidence provided by multisensory information. The multisensory condition received overwhelmingly positive remarks, with participants particularly valuing the haptic feedback (vibrations). Conversely, participants disliked instances of information overload, irrelevant details, and poorly timed cues, expressing a clear preference for dictating what information they received, when it was delivered, and how it was presented. This desire manifested in requests for features like the

ability to toggle specific pieces of information on or off. As P2 articulated, users wanted control over options: *"I would say like every single thing that could possibly give you information, make sure that there's an on and off switch for every single piece. So that way, you can customize it a little bit more to your taste."* This feedback represents a desire for users to independently repeat and control audio cues, fostering greater autonomy in how and when support is accessed.

5 Discussion

Aggregate results indicated that the multisensory delivery prompted faster task completion, increased accuracy, and was preferred compared to the NL condition. Participants found that while the VAM was new and took some time to familiarize themselves with, it was a helpful tool for orienting during entry. The time to learn new haptic interfaces is a common finding across the literature [7, 13] and research shows that once learned, haptic displays can significantly improve users' ability to perceive spatial structure and orientation on touch surfaces [21]. Additionally, research on multimodal learning shows that while relying on more than a single means of information supports more efficient cognitive processing [23] these redundant cues are most effective when they are structured and complementary rather than overwhelming [22]. This aligns with how, after practice, the two forms of delivery used here (haptics and audio) acted as mutual reinforcers, allowing orientation and reorientation in meaningful ways. Participants who had identified and sat in the incorrect seat during the NL condition noted that it was challenging to orient themselves with the audio-only descriptions. For instance, statements such as "left most seat" are difficult to confidently navigate toward without a map to indicate its location, further supporting that multisensory information provides improvements in environmental perception compared to NL information [5]. The seat finding time results suggest that multisensory information could improve seat localization for BLV travelers, especially in crowded AVs. As AVs become more prevalent, the ability for BLV travelers to safely and quickly find their seat is essential to ensuring ease of use and timeliness of travel.

While the detailed information provided in the multisensory condition was often useful, participants felt it was not always necessary. Nearly all participants expressed wanting to adjust the amount and modality of the information delivered, such as by using device-native accessibility settings (e.g., text-to-speech, VoiceOver). This aligns with prior work demonstrating the importance of BLV user control in AVs [2, 4]. Using personal devices with individually customized settings was also suggested as a means to adjust the speed of the audio information. Participants suggested numerous other specific customization implementations as well, including confirmations of reaching the correct seat or vehicle, detailed seat maps (e.g., orientation, direction, seating layout), and adaptive sound levels.

However, in the multisensory condition, every individual correctly found their seat, demonstrating the advantage of more perceptual, spatial cues in providing clear guidance that is not dependent on cognitive processing of instructions. This echoes evidence that perceptual cues often outperform verbal instructions for layout and orientation tasks [1, 10]. Taken together, the results presented here build on existing work exploring accessibility during the entry portion of the trip with AVs [9], suggesting the important role that new multisensory tools can play, while challenging assertions that audio-only interactions are preferred by BLV travelers.

6 Conclusion and Future Work

The insights from this study add to the growing corpus of work suggesting BLV users benefit from multisensory tools but desire greater agency over the information they receive, including the ability to request information on-demand and adjust the timing and amount of detail provided. Future research should focus on extending these results to on-road testing with more participants and fully developed apps for higher ecological validity. Investigating how to best implement these controls would be a valuable next step. Future work could also expand the content of the informational cues in the description logic tested here to include new categories, such as real-time traffic updates or details about specific landmarks, and mapping these to a richer vocabulary of haptic and spatialized audio patterns. In so doing, the multisensory interfaces developed can continue to be optimized for BLV user needs, promoting inclusive transportation systems leveraging AVs in the future.

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